

MINI PROJECT REPORT ON
3D CAD Modelling of a Farmer Assistant Robot

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**VASAVI COLLEGE OF ENGINEERING (AUTONOMOUS)
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CERTIFICATE

This is to certify that the mini project report entitled “**3D CAD Modelling of a Farmer Assistant Robot**” is a bonafide record of the mini project carried out by

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in the Department of Mechanical Engineering, Vasavi College of Engineering (Autonomous) (affiliated to Osmania University), Hyderabad during the academic year 2025-26 in partial fulfillment for the award of Degree in Bachelor of Mechanical Engineering.

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Abstract

The rapid growth of global food demand and the increasing challenges faced by farmers—such as labour shortages, unpredictable climate conditions, and the spread of crop diseases—have accelerated the adoption of smart agricultural technologies. This mini-project focuses on the 3D modelling and conceptual design of a Farmer Assistant Robot, developed to support farmers in efficient field monitoring and early plant disease detection. The robot is modelled as a compact, all-terrain, tracked vehicle capable of navigating through farmland with stability and ease. Its dual-track mobility system ensures reliable movement across uneven soil surfaces, while the articulated chassis enhances maneuverability during field operations.

A key feature of the design is the integrated solar panel, mounted on an adjustable support arm to maximize energy harvesting during operation. This renewable-energy approach not only extends the robot's working duration but also makes it suitable for deployment in large agricultural fields with limited access to charging infrastructure. The central body of the robot includes dedicated compartments for housing imaging sensors, camera modules, and processing electronics responsible for analyzing plant leaves and identifying early symptoms of disease through image-based detection techniques.

The objective of the project is to demonstrate how a cost-effective robotic platform can reduce manual monitoring efforts and improve the accuracy and timeliness of disease identification. By enabling continuous field inspection and delivering real-time diagnostic insights, the robot serves as a valuable tool for promoting healthier crop growth, reducing yield loss, and supporting precision agriculture practices. This 3D model lays the foundation for developing an operational prototype that can assist farmers in transitioning toward more automated, sustainable, and data-driven farming systems.

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Chapter 1

Introduction

1.1 Background

Agriculture is one of the most essential sectors in India, yet farmers continue to face challenges such as labour shortages, time-consuming manual operations, and low efficiency in repetitive tasks. With the growing need for improved productivity and sustainable farming practices, modern technologies such as automation, robotics, and CAD-based design are increasingly being adopted in agriculture. Robots designed for assisting farmers can significantly reduce physical effort and improve accuracy in field operations.

1.2 Need for Farmer Assistant Robots

Traditional farming relies heavily on manual labour, which is becoming increasingly scarce and expensive. Many essential tasks—such as spraying, soil monitoring, material transportation, and crop handling—demand precision and consistent performance. Farmer Assistant Robots offer a practical solution by performing such tasks efficiently with minimal human involvement. They enhance productivity, ensure safety in hazardous operations like pesticide spraying, and contribute to precision farming practices that save both time and resources.

1.3 Role of 3D CAD Modelling

3D CAD modelling plays a crucial role in the development of modern robotic systems. It enables detailed visualisation, accurate measurement, assembly planning, and motion simulation before the robot is physically built. Through CAD tools, individual components can be modelled and optimised to ensure proper fit and function. This reduces design errors, saves prototyping costs, and provides a strong foundation for fabrication and future automation integration.

1.4 Project Purpose

The objective of this project is to create a comprehensive 3D CAD model of a Farmer Assistant Robot, including its chassis, wheel system, robotic arm or tool mechanism, and sensor or equipment mounts. This model serves as a conceptual design that demonstrates how a robot can support farmers in simple field activities.

1.5 Scope of the Work

The project is limited to mechanical design and 3D CAD modelling. It does not include manufacturing, electronics, control systems, or field testing.

CHAPTER- 2

LITERATURE REVIEW

The literature reviewed proved highly valuable, offering concepts and techniques that directly supported the development of our project.

1. Fast and Accurate Detection and Classification of Plant Diseases

The proposed software enhances an existing method for automatically detecting and classifying plant leaf diseases by improving both speed and accuracy. After segmentation, it adds two steps: identifying mostly green pixels and masking them using Otsu-derived thresholds, followed by removing zero-RGB pixels and boundary pixels from infected regions. These refinements produce a more robust detection process. Experimental results show the system achieves 83–94% classification precision and delivers a 20% speedup over the earlier approach.

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3. Design and development of an automated and non-contact sensing system for continuous monitoring of plant health and growth:-

An automated system was designed and built to continuously monitor plant health and growth in a controlled environment using a distributed system approach for operational control and data collection. The computer-controlled system consisted of a motorized turntable to present the plants to the stationary sensors and reduce microclimate variability among the plants. Major sensing capabilities of the system included machine vision, infrared thermometry, time domain reflectometry, and micro-lysimeters. The system also maintained precise growth-medium moisture levels through a computer-controlled drip irrigation system. The system was capable of collecting required data continuously to monitor and to evaluate the plant health and growth.

4. Monitoring Plant Health Using-Driven IoT Imaging system :

The early detection of plant health issues is essential for maximizing crop yield, reducing resource waste, and ensuring sustainable farming. Traditional methods of crop monitoring are labor-intensive, error-prone, and often reactive rather than Proactive. This article explores how AI-driven IoT Imaging systems are transforming plant health monitoring by enabling real-time, automated, and Highly accurate diagnosis of crop stress, nutrient deficiencies, pests, and diseases. These systems use Multispectral and thermal imaging sensors connected to cloud platforms and artificial intelligence Models to detect anomalies in plant physiology long before visible symptoms appear. The integration Of IoT and AI not only enhances the precision and speed of plant health assessments but also supports Smarter decision-making in precision agriculture.

5. Design of a Plant Health Monitoring System for Enhancing Food Safety of Space Crop Production Systems

A prototype hyperspectral and chlorophyll fluorescence imaging system was developed to enable early, automated detection of plant stress in space-based crop production, avoiding delays inherent in monitoring leaf-area growth. The system includes a hyperspectral spectrometer, LED light banks, and a translational arm that scans plants to capture reflectance and fluorescence signals. Processed images allow background removal, reflectance calculation, and vegetation-index generation. Future work will refine fluorescence imaging and identify indices for detecting stresses such as drought, nutrient imbalance, and fungal infection.

6. Prediction and Analysis of Paddy Crops Disease in Artificial Intelligence Techniques-

Agricultural productivity is vital for human survival, and paddy is a key crop supporting national economic growth. However, its yield is often reduced by diseases caused by bacteria, fungi, viruses, and harmful insects, especially when infection occurs early. Traditionally, farmers detected diseases manually, but this approach is slow and error-prone. Image processing has emerged as an effective solution, enabling accurate identification and classification of plant diseases, weed discrimination, and disease forecasting, thereby improving crop management and cultivation efficiency.

CHAPTER- 3

Problem Statement:-

- Farmers struggle to detect plant diseases early.
- Checking each plant manually takes a lot of time and effort.
- Early symptoms on leaves are small and easy to miss.
- Delayed detection leads to big crop losses.
- Many farmers don't have quick access to experts for diagnosis.
- Wrong or late treatment increases cost and reduces yield.

Objective of the Project:-

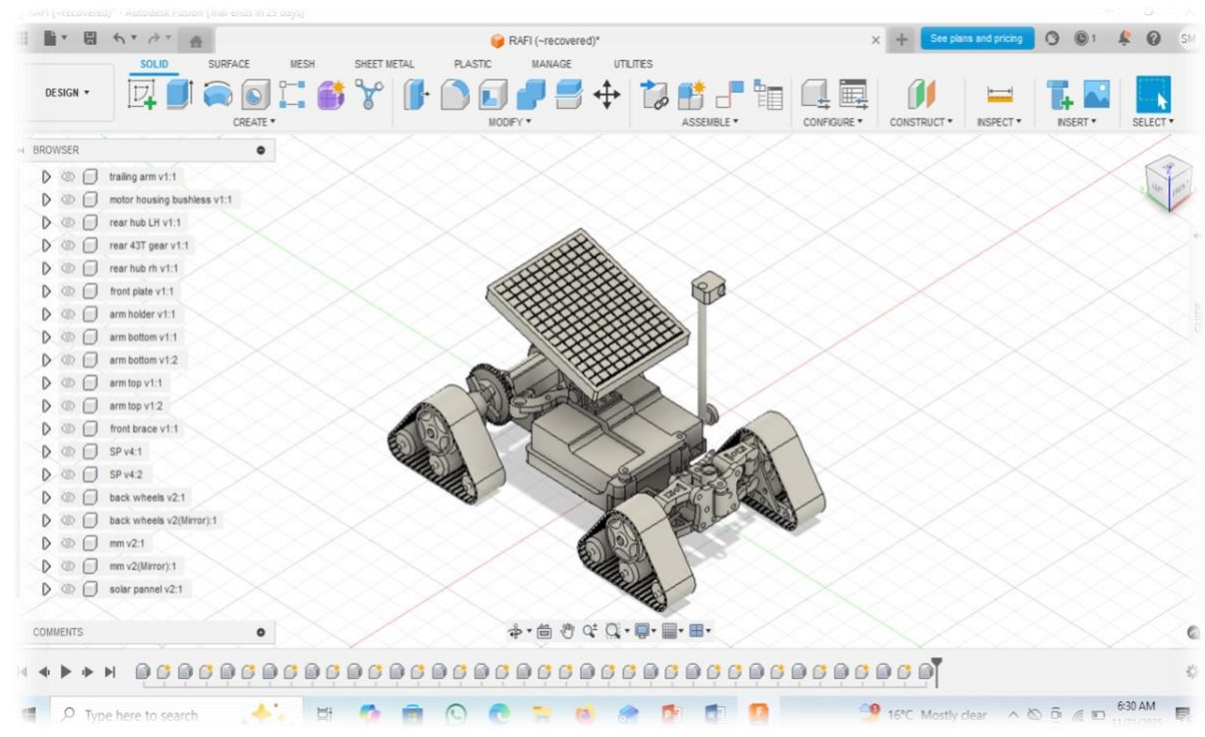
- Design a robotic assistant that can automatically monitor plants and detect early signs of leaf disease.
- Capture clear images of plant leaves using an onboard camera and analyze them using a detection algorithm.
- Provide quick alerts to farmers so they can take timely action and prevent disease spread.
- Reduce manual inspection, save time and labor, and support precision farming.
- Build a field-friendly robot that can move easily and work effectively in outdoor conditions.

Outcomes:-

- A CAD model of a robot for plant-leaf disease detection was successfully designed
- The model clearly shows the structure, components, and layout for the CNN-based detection system .
- The final CAD design is stable, easy to assemble, and suitable for real-world implementation.

Chapter 4

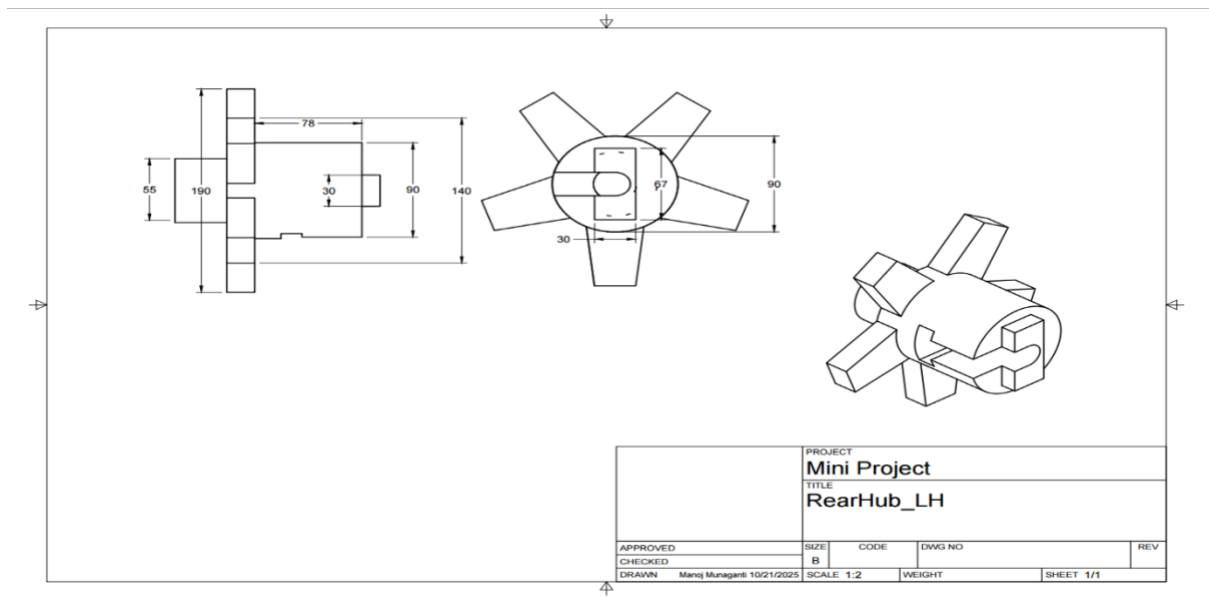
Cad model of the Robot



This image shows our robot model created in Fusion 360. It includes a main body, track wheels systems for movement, support arms, and a solar panel mounted on top for power. The design helps us understand how each part connects and works together. We added this image to our report to clearly present the robot's structure and layout.

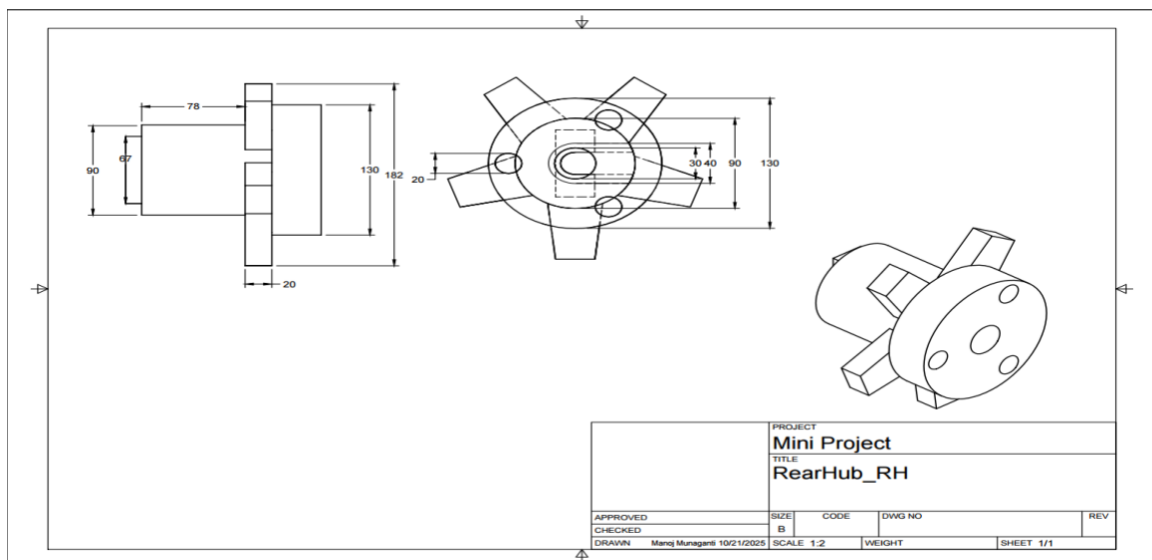
Components of the robot

1.



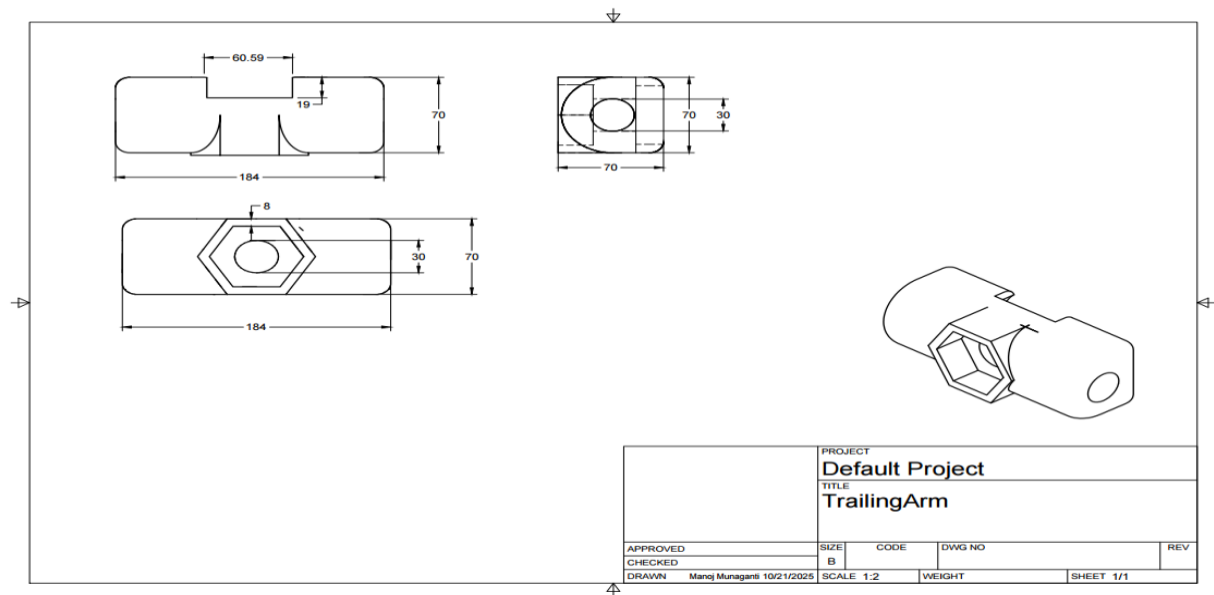
Rear Hub left hand side - The left side of the rear hub helps hold the axle in position and keeps the wheel centered in the frame.

2.



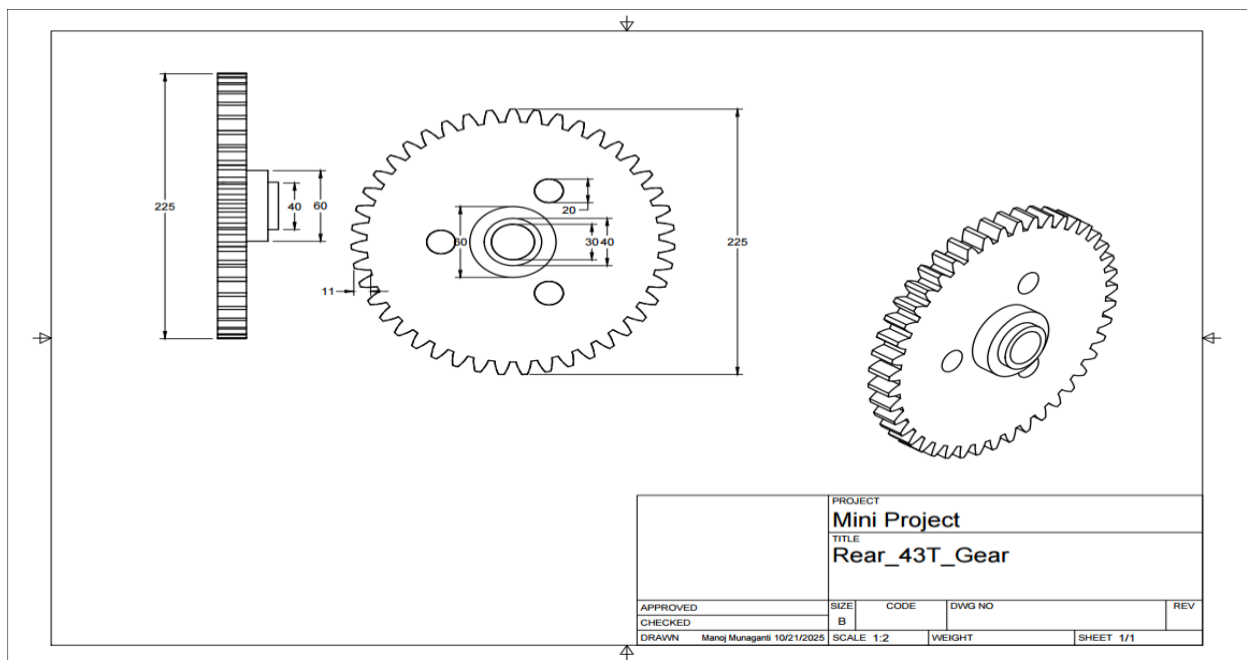
Rear hub right hand side : The right side of the rear hub helps hold the axle in position and keeps the wheel centered in the frame.

3.



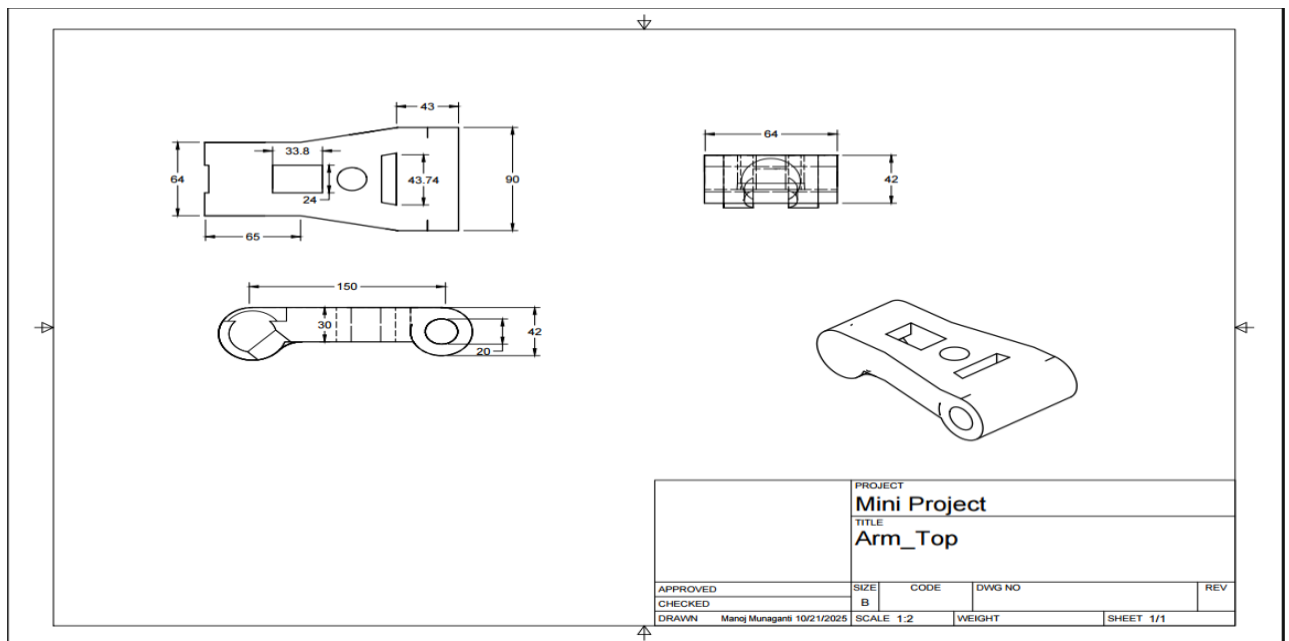
Trailing Arm : Connects both Rear hub right hand and left hand side ,
transmitting motion and load.

4.



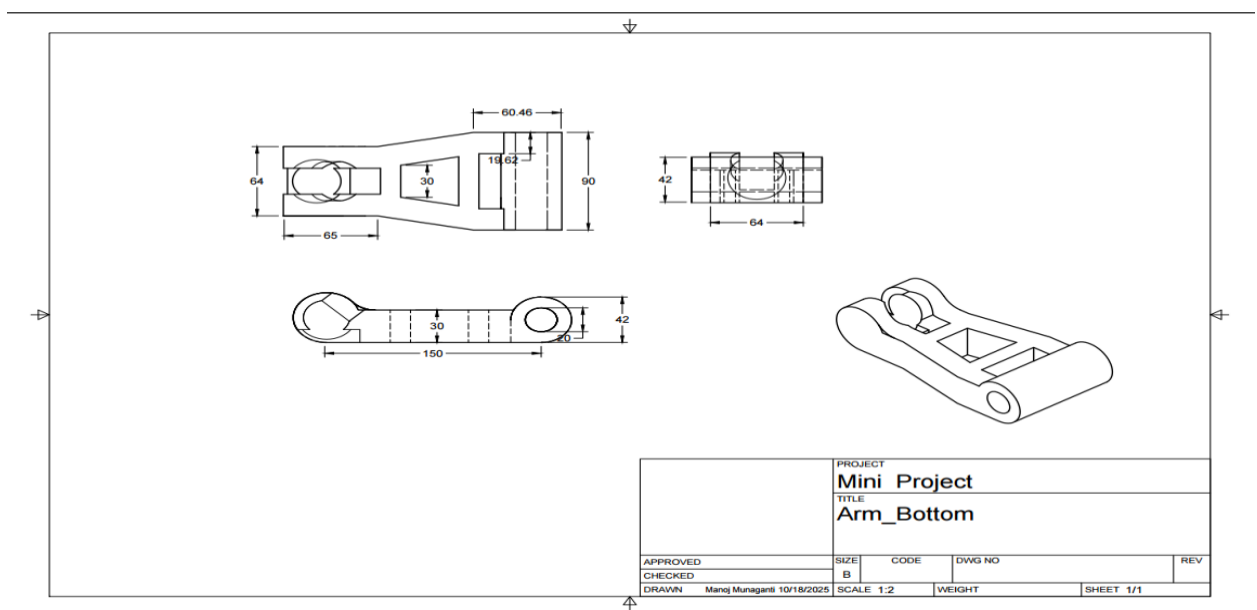
Gear : The gear receives rotation motion and turns the rear wheel.

5.



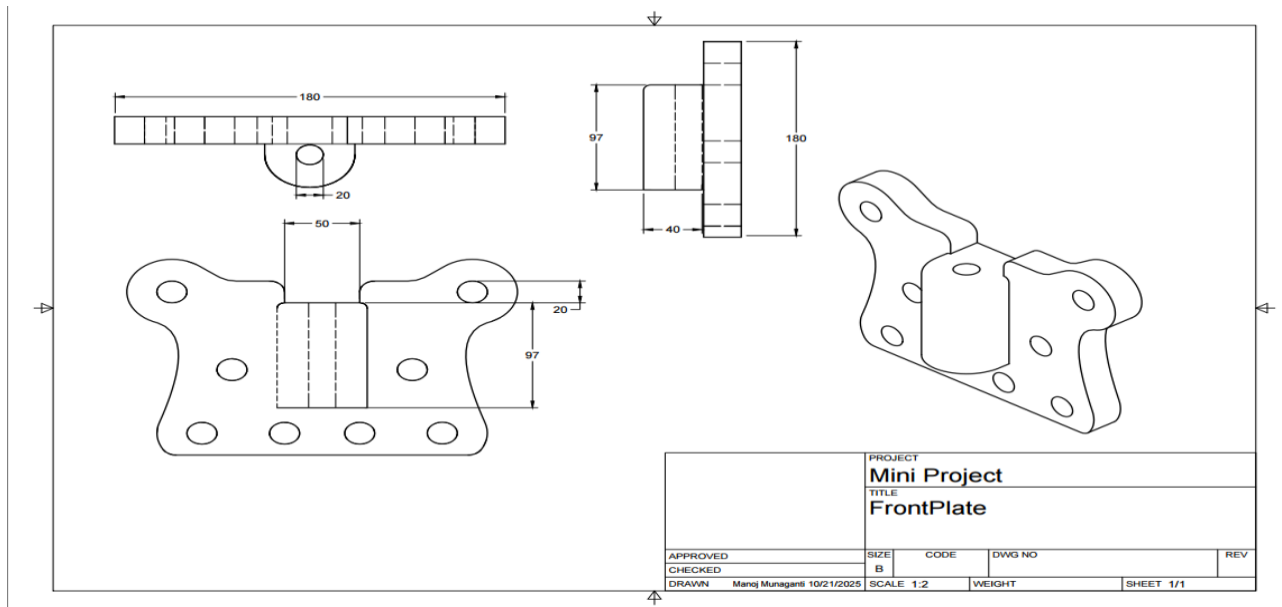
Arm top : The arm top connects to front plate which creates the suspension for the front portions of rover which is the function of the arm top.

6.



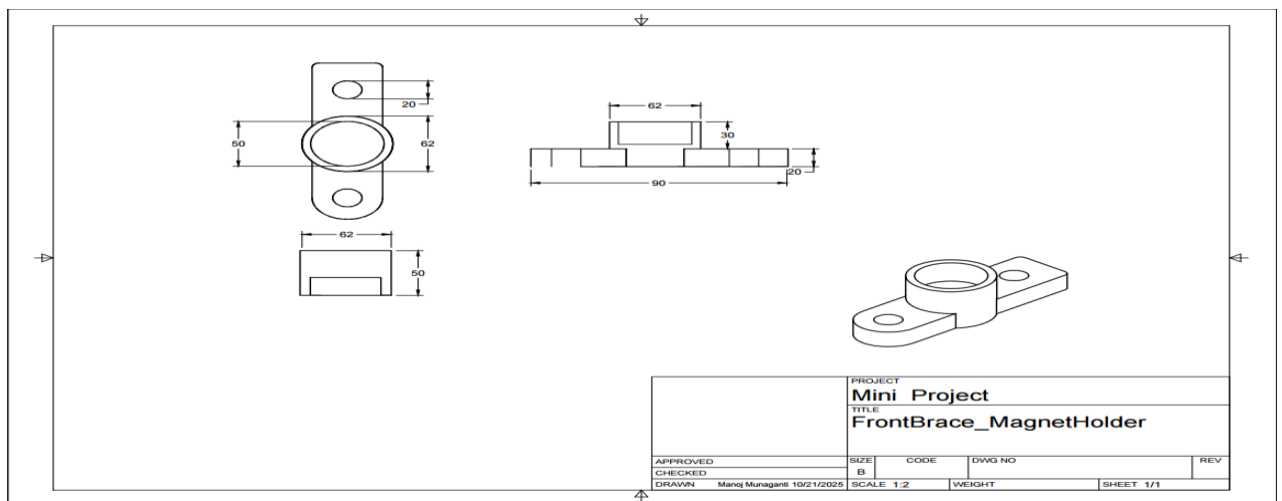
Arm bottom : The arm bottom connects to front plate which creates the suspension for the front portions of rover.

7.



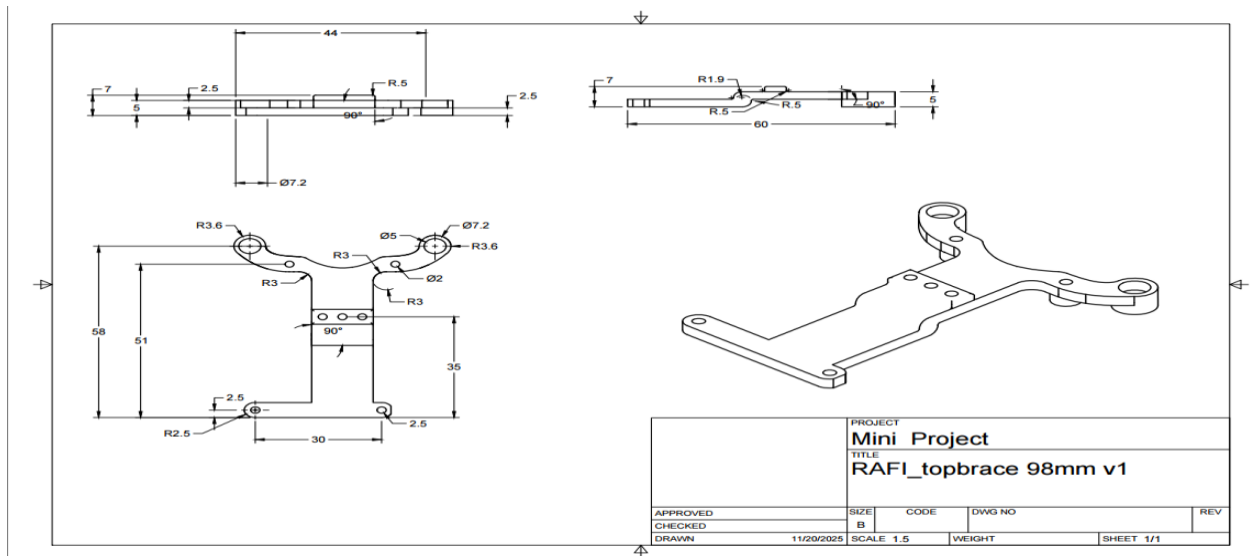
Front Brace : The front Brace holds the two front plates as per required suspension

8.



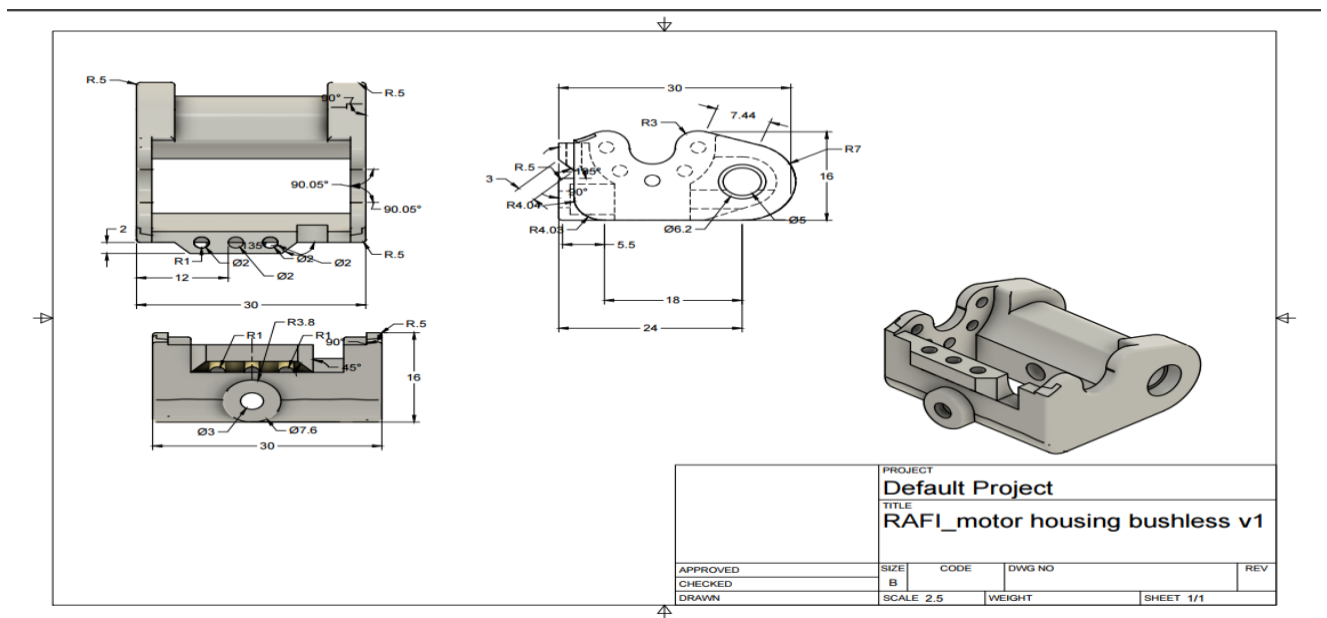
Front Brace : The front Brace holds the two front plates as per required suspension

9.



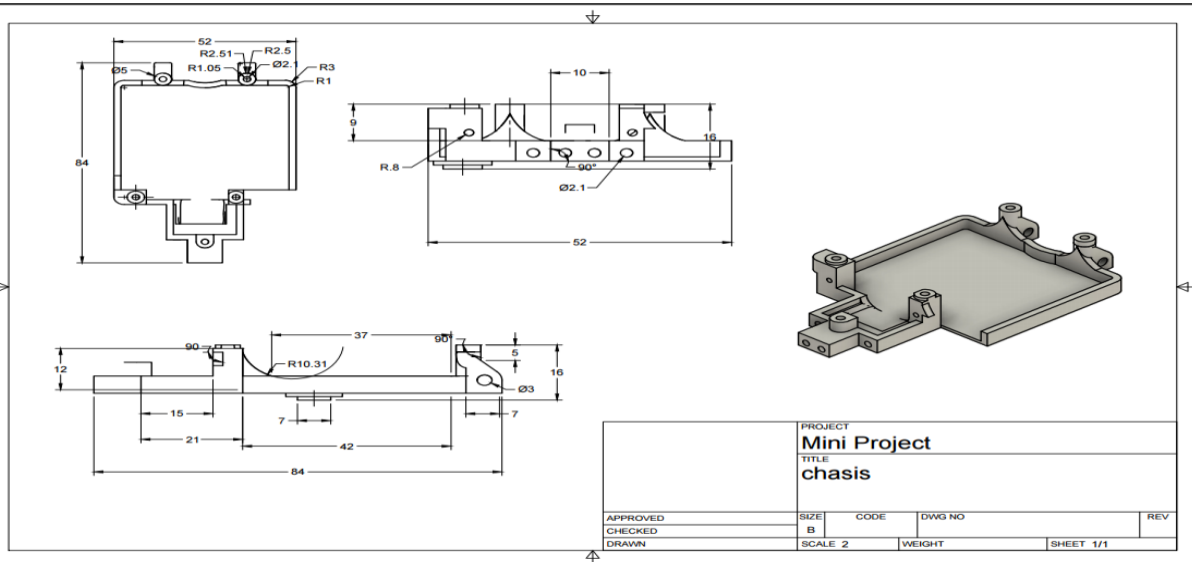
Top brace : It's function is to cover the components which are inside the chasis

10.



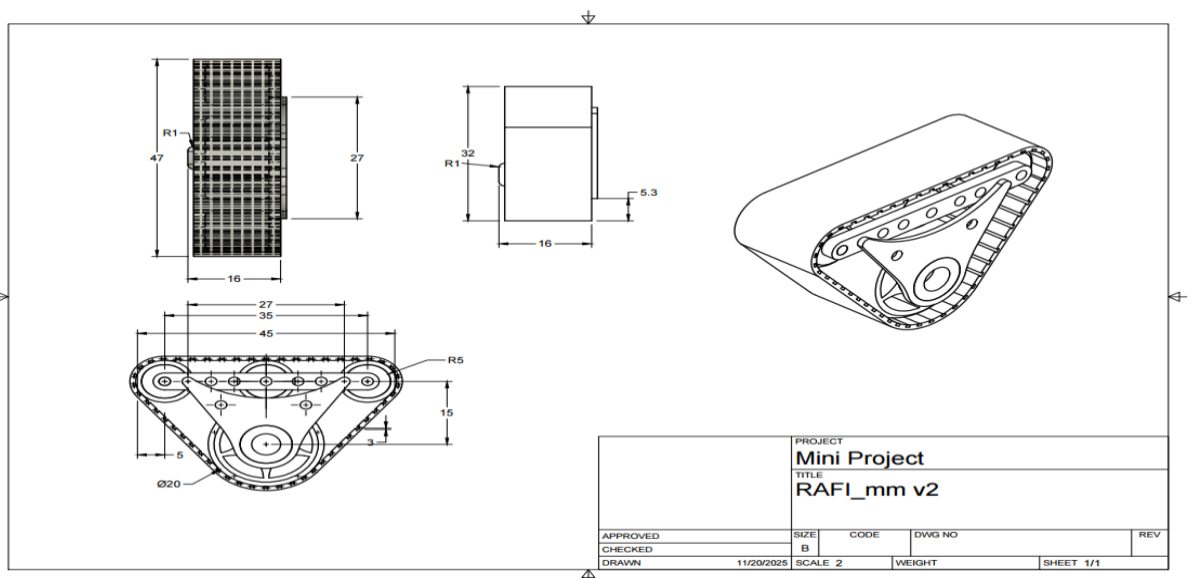
Motor Housing : The motor housing name it self says that it protects motor modules , it connects to the gear present in left hand side

11.



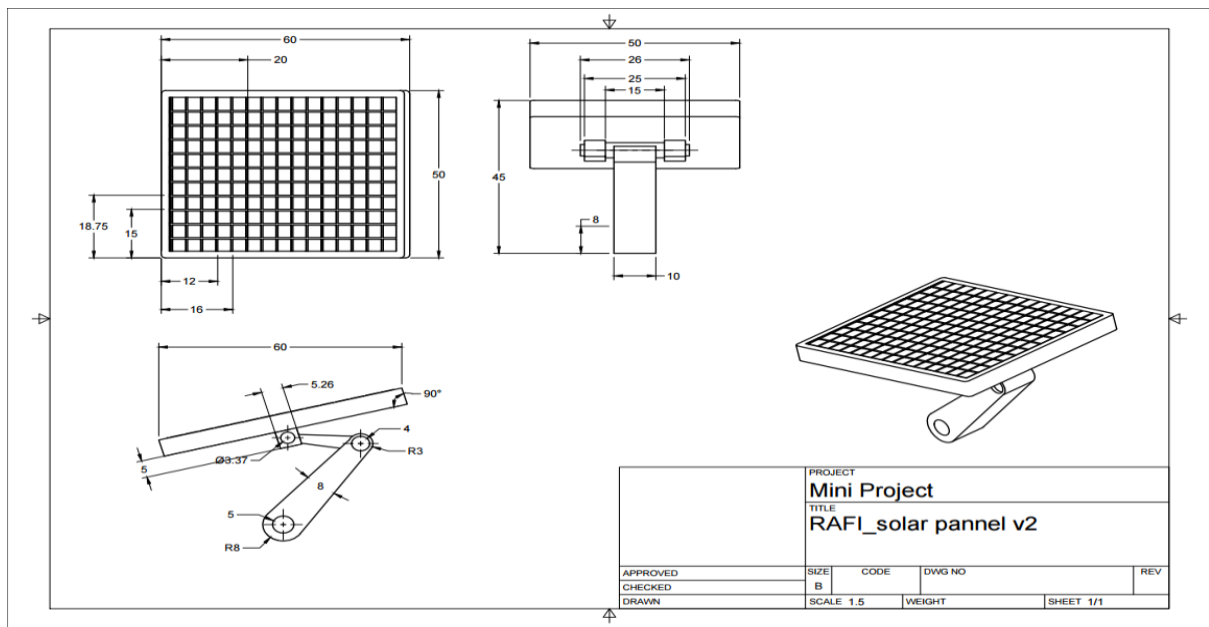
Chasis :- It holds all the components mounted on it. It acts as a connection between front and back wheels.

12.



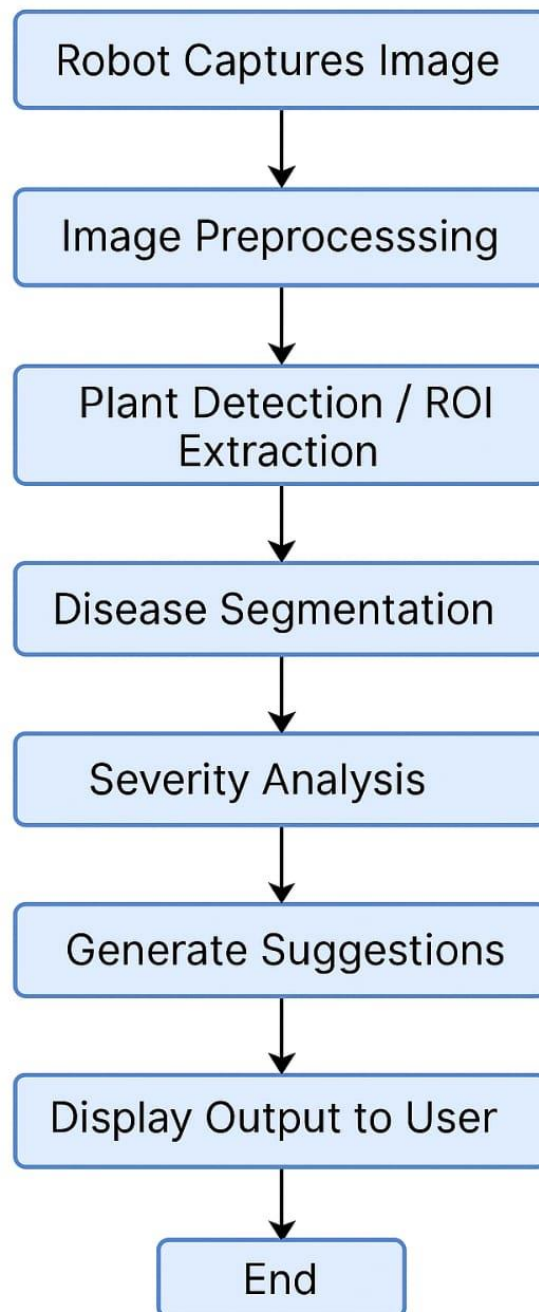
Track wheel : Its function is to keep the track in place and let it move smoothly and also to distribute pressure and reducing soil compaction.

13.



Solar pannel : A solar panel converts sunlight into electrical energy by using photovoltaic cells .

Methodology



The robot begins by capturing images of the plant in the field and improves the clarity of these images through preprocessing. It then identifies the plant or the important region within the image and extracts it for further study. Using segmentation techniques, the robot detects any diseased portions of the plant and analyzes how severe the infection is by measuring the affected area. Based on this analysis, it automatically generates suitable suggestions or recommendations for treatment. The final results, including detected disease and suggested actions, are then displayed to the user in an easy-to-understand format.



Chapter 5

Results :-

In this project, a complete 3D model of a Farmer Assistant Robot was successfully created. The final model includes the main parts needed for an agricultural robot, such as: Tracked wheels, which help the robot move smoothly on soil and uneven farmland. A solar panel, mounted on an adjustable stand, which provides renewable power for longer operation. Sensor and camera mounts, where plant disease detection tools can be placed. A compact body design, which allows easy movement and space for electronic components. The 3D model clearly shows how all these parts fit together to form a fully functional robot design.

Discussion :-

The results show that the robot design is practical and suitable for real agricultural use. The tracked movement system makes the robot more stable and better suited for farm conditions compared to normal wheels. This means it can move between crops without getting stuck easily. The solar panel is an important feature because it allows the robot to work for long hours in the field without needing frequent charging. This makes the robot more useful for farmers who work in large fields. The space provided for sensors and cameras makes it possible for the robot to detect plant diseases early. Early detection helps farmers take action quickly, reducing crop damage and improving productivity. Overall, the simple 3D model proves that such a robot can help farmers by reducing manual work, improving plant monitoring, and supporting healthier crop growth.

Chapter 6

Conclusions:-

The robotic assistant equipped with a Convolutional Neural Network (CNN) is able to detect plant leaf diseases quickly, accurately, and at an early stage. The CNN model improves accuracy by identifying even small spots, color changes, or patterns that humans may miss.

This technology promotes precision farming, reduces unnecessary pesticide use, and ultimately increases crop productivity. Overall, the robot provides a smart, reliable, and efficient solution to support farmers and improve agricultural health management.

FUTURE SCOPE :-

The Farmer Assistant Robot designed in this project has strong potential for further development and real-world application. In the future, several improvements and additions can make the robot even more useful for farmers:

1. Real Prototype Development

The next step is to convert the 3D model into a working physical prototype to test the robot in actual farmland conditions.

2. Advanced Disease Detection

More accurate sensors and AI-based image processing can be added to identify a wider range of plant diseases with higher accuracy.

3. Wireless Data Transfer

Features like Wi-Fi, Bluetooth, or IOT modules can allow the robot to send real-time data to farmers' mobile phones or computers.

4. Automatic Navigation

GPS and obstacle detection sensors can be integrated to help the robot move independently through the fields without human control.

5. Multi-Purpose Attachments

Tools like small sprayers, soil moisture sensors, or robotic arms can be added to perform extra tasks such as watering plants, applying pesticides, or picking samples.

6. Battery Backup System

Along with the solar panel, an improved battery system can help the robot work even during cloudy weather or at night.

7. Scaling for Large Farms

The design can be upgraded to cover larger fields or adapted into multiple sizes depending on different types of farming.

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